Customer Stories / Professional Services / New Zealand

2023

Actionstep Manages 400% More Traffic on Legal Platform with Amazon Aurora Serverless

Find out how the international legal practice management software provider boosted scalability, performance, and the customer experience by migrating to a serverless infrastructure.

Overview | Opportunity | Solution | Outcome | AWS Services Used

Up to 400%

Increase in traffic handled without disruptions during peak periods

100%

Reduction in the number of manually

Up to 10 hours/week

Man hours saved by using a fully managed relational database

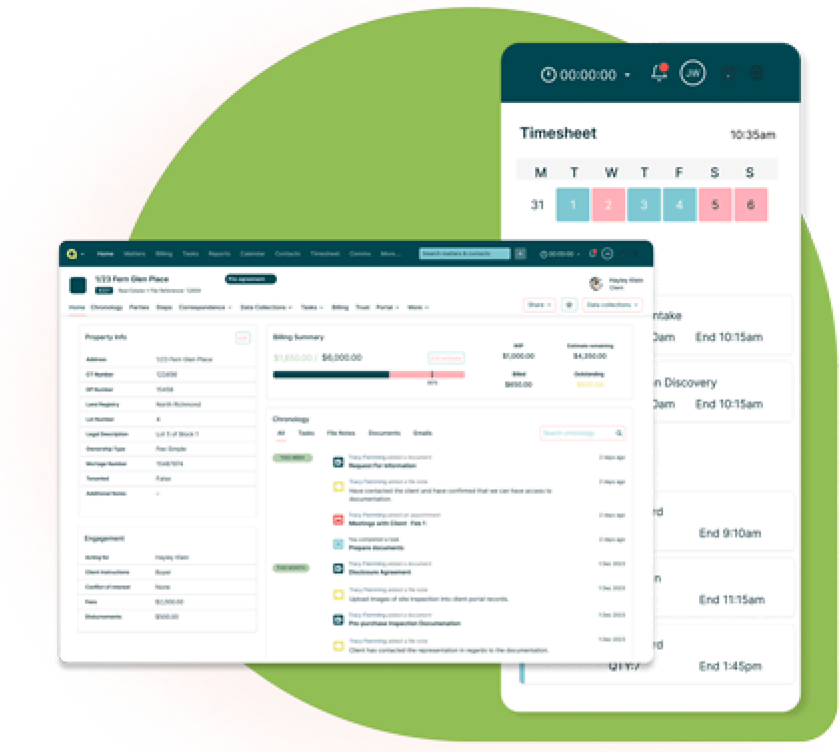
> **Zero downtime**
>
> Since migrating to a serverless infrastructure on the AWS Cloud

Overview

The legal industry is heavily reliant on paperwork, from contracts to billing documents, which poses challenges for legal professionals who must navigate detailed creation and review processes, while managing client relationships and delivery timelines. This detailed, skilled work has no room for errors and delays.

Actionstep is a cloud-based legal practice management platform that aims to simplify routine legal and administrative processes for law firms. The company uses automation to handle workflow processes including document assembly and management, cataloguing, and billing. Founded in 2004, Actionstep now serves over 35,000 users worldwide, including Australia, New Zealand, the United Kingdom, and the United States.

Built on Amazon Web Services (AWS), the company previously used Amazon Relational Database Service for PostgreSQL (Amazon RDS for PostgreSQL), a fully managed relational database, to store critical data, such as client communications and billing information. In 2023, Actionstep began migrating to Amazon Aurora Serverless v2, an on-demand, autoscaling configuration for Amazon Aurora, to better match its evolving business needs. After the migration, the company now has a database platform that can accommodate unpredictable workloads and effectively scale to manage sudden spikes in activity, all while increasing workflow efficiency for end customers.



Opportunity | Using AWS to Automate Scalability and Optimize Performance

Before migrating to a serverless infrastructure, Actionstep's platform managed an average of 25,000 active users and hundreds of millions of queries daily. During peak times, traffic would surge by up to 400 percent within a single week, requiring extensive pre-planning for additional computing resources. Despite the efforts, the company still had to deal with challenges such as slower database response times and diminished performance due to workload unpredictability. Actionstep estimated that managing scaling and rectifying instance sizing in a multi-tenant environment to match peaks would require manual efforts which took away time otherwise spent on engineering activities.

Furthermore, the company had to put staff in charge of overseeing and managing 55 production database clusters for thousands of global customers. Even during non-peak periods, this demanding task required an average of up to 10 working hours each week.

Actionstep recognized that adopting a serverless infrastructure would allow the company to seamlessly scale workloads and meet real-time demands as its customer base grows. A serverless infrastructure would also allow Actionstep to adopt a variety of cutting-edge solutions.

Solution | Enhancing User Experience with a Serverless Database Transformation

In February 2023, Actionstep began migrating 36 terabytes worth of data from Amazon RDS for PostgreSQL to Amazon Aurora Serverless v2, completing the move by October 2023—two months ahead of schedule. The platform upgrade removes much of the complexity of managing database instances and capacities, freeing the staff to focus on high-value tasks. For instance, the team can now prioritize working on customer-centric features.

More importantly, Actionstep now has the agility to scale flexibly and reliably to meet growing workload demands seamlessly. With Amazon Aurora Serverless v2, Actionstep no longer has to worry about tenan oned capacity limits and causing performance issues. With in-place scaling offered by the Aurora pl ep can now better support its customers, thus allowing legal professionals to retrieve documents, information, and perform tasks quickly and reliably. This has led to enhanced efficiency and productivity. Furthermore, a fast database minimizes the risk of working with outdated or inaccurate data, reducing the likelihood of errors and ensuring that legal practitioners can confidently rely on the information at hand.

AWS played a pivotal role in facilitating the seamless migration and providing training workshops and playbooks. AWS also helped Actionstep upgrade its PostgreSQL version 12 to version 13 with minimal service disruption. The team then used the RDS platform to seamlessly upgrade the database engine version in place to PostgreSQL v15 to take advantage of new optimizer features including improved sort performance. Actionstep also used Aurora Read Replicas to perform real-time data migration from RDS while keeping existing systems running and successfully cut over to the serverless infrastructure, minimizing downtime to just a few hours.

Outcome | Fueling Innovation and Growing the Customer Base

Actionstep's serverless infrastructure gives the company the flexibility to add tiers to its platform offerings, with the aim of providing more data to customers where they need it. The company also plans to set up a data warehouse, allowing it to further expand on offering insights on data.

As Actionstep continues to expand its midsize customer market to firms up to 400 staff, the serverless infrastructure provides unlimited opportunities for scale and expansion.

“By working with AWS, we gained access to industry experts with deep knowledge about database migration and modernization. These experts gave us the confidence to execute a full migration to a serverless infrastructure. Along with the ability to flexibly scale our operations, we are now in a stronger position to expand our presence in new markets,” said Stevie Mayhew, vice president of engineering at Actionstep.

Learn More

To learn more, visit aws.amazon.com/smart-business.

About Actionstep

Actionstep is an international legal practice management platform provider that aims to simplify document management for legal firms. Its platform is a one-stop shop for billing, reporting, document management, and customer relationships management, among others. Since its founding in 2004, Actionstep has amassed over 35,000 users across thousands of practices around the world, including Australia, New Zealand, the UK, and the US.

AWS Services Used

Amazon RDS for PostgreSQL

Amazon RDS makes it easier to set up, operate, and scale PostgreSQL deployments on the cloud. With Amazon RDS, you can deploy scalable PostgreSQL deployments in minutes with cost-efficient and resizable hardware capacity.

Learn more »

Amazon Aurora Serverless v2

Amazon Aurora Serverless is an on-demand, autoscaling configuration for Amazon Aurora. It automatically starts up, shuts down, and scales capacity up or down based on your application's needs. You can run your database in the cloud without managing any database instances. You can also use Aurora Serverless v2 instances along with provisioned instances in your existing or new database clusters.

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